26. Einstein frequently maintained that the experiments of Michelson and Morley were of no importance in allowing him to arrive at special relativity. I believe this to be true, and that it illustrates an important factor in the philosophy of science. In order to make advances in our understanding of the world, it is not always necessary to have new data. Copernicus had no more observational data than Ptolemy: he was able to deduce heliocentrism from the data available to Ptolemy by interpreting it better—as Einstein did with regard to Maxwell.

27. If I see my sister through a telescope celebrating her twentieth birthday and send her a radio message that will arrive on her twenty-eighth birthday, I can say that *now* is her twenty-fourth birthday: halfway between when the light departed from there (20) and when it returned (28). It's a nice idea (not mine: it's Einstein's definition of "simultaneity"). But this does not define a common time. If *Proxima b* is moving away, and my sister uses the same logic to calculate the moment simultaneous to her twenty-fourth birthday, she *does not* obtain the present moment here. In other words, in this way of defining simultaneity, if for me a moment A in her life is simultaneous with a moment B in mine, the contrary is not the case: for her, A and B are not simultaneous. Our different speeds define different surfaces of simultaneity. Not even in this way do we obtain a notion of a common "present."

28. The combination of events that are at spacelike distance from here.

29. Among the first to realize this was Kurt Gödel; see "An Example of a New Type of Cosmological Solutions of Einstein's Field Equations of Gravitation," *Reviews of Modern Physics* 21 (1949): 447–50. In his own words: "The notion of 'now' is nothing more than a certain relation between a certain observer and the rest of the universe."

30. Transitive.

31. Even the existence of a relation of partial order might be too strong with regard to reality, if closed temporal curves exist. On this subject see, for example, Marc Lachièze-Rey, *Voyager dans le temp: La Physique moderne et la temporalité*, (Paris: Editions du Seuil, 2013).

32. The fact that there is nothing logically impossible about travels to the past is demonstrated clearly in an engaging article by one of the great philosophers of the last century: David Lewis, "The Paradoxes of Time Travel," *American Philosophical Quarterly* 13 (1976): 145–52, reprinted in *The Philosophy of Time*, eds. R. Le Poidevin and M. MacBeath (Oxford: Oxford University Press, 1993).

33. This is the representation of the causal structure of a black hole metric in Eddington-Finkelstein coordinates.

34. Among the dissenting voices, there are those of two great scientists for whom I have a particular friendship, affection, and admiration: Lee Smolin (*Time Reborn* [Boston: Houghton Mifflin Harcourt, 2013]) and George Ellis ("*On the Flow of Time*," Fqxi Essay, 2008, https://arxiv.org/abs/0812.0240; "The Evolving Block Universe and the Meshing Together of Times," *Annals of the New York Academy of Sciences* 1326 [2014]: 26–41; *How Can Physics Underlie the Mind?* [Berlin: Springer, 2016]). Both insist that there must exist a privileged time and a real present, even if these are not captured by current physics. Science is like affection: those who are dearest to us are those with whom we have the liveliest disagreements. An articulate defense of the fundamental aspect of the reality of time can be found in Roberto M. Unger and Lee Smolin, *The Singular Universe and the Reality of Time* (Cambridge, UK: Cambridge University Press, 2015). The point of view of Smolin and Ellis is defensible. But is it fruitful? The choice is between forcing the description of the world so that it adapts to our intuition, or learning instead to adapt our intuition to what we have discovered about the world. I have few doubts that the second strategy is the most fruitful one.